

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-201

WASTE DISCHARGE REQUIREMENTS
FOR
YOSEMITE VISTA ESTATES MUTUAL WATER AND SANITATION COMPANY
NOVASEL AND SCHWARTE INVESTMENT INC.
YOSEMITE VISTA ESTATES
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The Yosemite Vista Estates Mutual Water and Sanitation Company and Novasel and Schwarte Investment Inc. (hereafter jointly referred to as Discharger) submitted a Report of Waste Discharge (RWD), dated 11 April 2000, to treat and dispose of domestic sewage generated at the Yosemite Vista Estates community.
2. The facility is on Assessor's Parcels Number 66-220-50 and 66-220-63, and is owned by Novasel and Schwarte Investment Inc. and operated by the Yosemite Vista Estates Mutual Water and Sanitation Company.
3. The wastewater treatment plant is located approximately 7.5 miles east of Groveland on the north side of Ferretti Road immediately adjacent to Indian Creek. Yosemite Vista Estates is in Section 20, T1S, R17E, MDB&M, as shown on Attachment A, which is attached hereto and made part of the Order by reference.
4. Order No. 85-076, adopted by the Board on 26 April 1985, prescribes requirements for package plant system and spray fields. This Order is neither adequate nor consistent with the current plans and policies of the Board.
5. Wastewater originates from domestic sources in a 28-acre residential subdivision including a clubhouse and pool. The flow rates are expected to increase over time with growth in the development. The current flow rate of 5,000 – 6,000 gallons/day is expected to increase to around 12,000 gallons/day by the year 2005 when 65 units are developed. The subdivision is projected to be completed by 2010 with 87 units occupied and then will produce an estimated flow rate of 19,000 gallons/day.
6. Water for domestic use is supplied by a well that is operated by the Discharger. The subdivision's well is rated at 60 gpm and includes a 120,000 gallon storage tank. The well is located on the south side of the development adjacent to Highland Drive South.
7. Sewage is conveyed from the residential sites to the treatment plant through a gravity collection system. The sewage collection system is owned and maintained by the Discharger. The RWD states that the influent wastewater strength for BOD is believed to range from 175 – 225 mg/l.

8. Treatment consists of an extended activated sludge package treatment plant and two percolation ponds. Disposal is currently accomplished by evaporation and percolation in the two ponds. The plant layout is presented in Attachment B, which is attached hereto and made part of the Order by reference.
9. The plant treatment units consist of an aeration tank, including contact chamber of 22,500 gallons, and a secondary clarifier of 6,000 gallons. A folic acid mixture is periodically added to help control filamentous bacteria and sludge bulking. The treatment plant is rated for a total daily flow of 21,500 gpd with an influent daily loading rate of 45 pounds of BOD.
10. The treatment plant is equipped with a high water alarm and a small containment pond to capture any overflows. The residents monitor the treatment plant visually on a daily basis. A certified operator or his agents perform the plant operations and maintenance twice a week.
11. Effluent from the activated sludge package treatment plant is pumped from a 750 gallon stilling well through a 2.5 inch force main into first of two percolation ponds. The ponds are connected by a channel and are operated in series fashion. The estimated storage capacity of the ponds is 400,000 gallons for the first pond and 570,000 gallons for the second pond. The ponds are located within 30 feet of Indian Creek.
12. The RWD and other information submitted by the Discharger shows the average effluent discharge concentrations are as follows:

<u>Constituent</u>	<u>Average Value</u>
Flow Rate	5,500 gpd
BOD	5.47 mg/l
SS	<0.1 ml/L

13. A 4 February 2000 inspection found that trees and other vegetation are growing on the pond dikes and within the ponds. Pond #1 has a significant amount of cattails growing in it. The Discharger states that it is currently utilizing Pond #1 as a wetlands treatment cell. The Discharger is in the process of clearing trees from the dikes in accordance with a staff directive, and proposes to continue utilization of Pond #1 as a wetland treatment cell by maintaining cattail growth in the pond in a controlled manner. This Order requires the submittal of an Operations and Maintenance Plan for the wastewater treatment plant and ponds that will describe procedures to operate and maintain the ponds in compliance with requirements and limitations specified in this Order.
14. The Discharger has installed a turbine pump in the channel between the ponds to transfer effluent to a third pond. The third pond is located on the south side of the development around 500 feet south of the drinking water well, as shown on Attachment B. The Discharger is considering the incorporation of this pond, along with an adjacent sprayfield,

into the treatment/disposal process at a future date to help accommodate growth should the capacity of the current pond system become exceeded. The pond and sprayfield had been authorized under Order 85-076. However, the Discharger has never utilized either the pond or the sprayfield. In addition, the Discharger did not include specifications for the use of the sprayfield with the April 2000 RWD. Therefore, this Order, although authorizing the use of the third pond and sprayfield for the disposal of wastewater, does so only after submittal of an engineering report and design plan, and approval of these reports by the Executive Officer.

15. The Discharger has recently installed a flow meter for monitoring effluent discharges from the treatment plant to the percolation ponds.
16. The majority of the waste sludge generated from the treatment plant is hauled to the Regional Wastewater Treatment Plant in Sonora for disposal. However, the Discharger also dries a small amount of sludge on site and applies it as compost to the land around the treatment plant. This Order requires a biosolids disposal plan and monitoring results if the Discharger wishes to continue applying sludge on site.
17. Federal regulations for stormwater discharges were promulgated by the U.S. Environmental Protection Agency on 16 November 1990 (40 CFR Parts 122, 123, and 124). The regulations require specific categories of facilities which discharge stormwater associated with industrial activities to obtain NPDES permits. The flow at this wastewater treatment plant is less than 1.0 mgd and therefore the Discharger is not required to apply for a stormwater NPDES permit.
18. Surrounding land uses are primarily residential and agricultural.
19. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
20. Surface water drainage is to Indian Creek, a tributary of Tuolumne River.
21. The beneficial uses of Tuolumne River from its source to New Don Pedro Reservoir are municipal and domestic supply; agricultural supply; hydropower generation; contact and non-contact recreation; warm and cold freshwater habitat; aesthetic enjoyment; groundwater recharge; fresh water replenishment; and preservation and enhancement of fish, wildlife, and other aquatic resources.
22. The beneficial uses of underlying groundwater are municipal, industrial, and agricultural supply.
23. The Board has considered anti-degradation pursuant to State Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to

determine whether the discharge will cause an increase in groundwater or surface water constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, line the ponds, implement source control, change the method of disposal, or take other action to prevent groundwater or surface water degradation.

24. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
25. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
26. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
27. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
28. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 85-076 is rescinded and the Yosemite Vista Estates Mutual Water and Sanitation Company and Novasel and Schwarte Investment Inc., its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Surfacing of wastewater outside and downgradient of the ponds is prohibited.
7. Discharge of untreated or partially treated waste to groundwater is prohibited.
8. The discharge of any wastewater other than that from domestic sources is prohibited.
9. Application of wastewater to areas other than those described in Findings No. 11 and 14 is prohibited.

B. Discharge Specifications:

1. The average dry weather discharge shall not exceed 19,000 gallons per day (gpd).
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. The wastewater treatment ponds shall not have a pH of less than 6.0 or greater than 8.5.
5. The Discharger shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.
6. The ponds shall be managed to prevent the breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the waste surface.

- b. Weeds shall be minimized through control of water depth, harvesting, and/or herbicides. If cattails are to be permitted to remain in Pond #1 for the purpose of effluent treatment and disposal, a letter of approval from the local mosquito abatement district must be submitted to the Executive Officer.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. The Discharger's wastewater treatment system shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
8. The freeboard in all ponds shall never be less than two feet as measured vertically from the water surface to the lowest point of overflow.
9. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.
10. On or about 1 October each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. 8 and No. 9.
11. The Discharger shall minimize impacts to the groundwater due to total dissolved salts to the maximum practical extent.

C. Effluent Limitations

The discharge of effluent in excess of the following limits is prohibited:

<u>Constituent</u>	<u>Units</u>	<u>30-Day Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40 ¹	80
Total Settleable Solids	ml/l-hr	0.5	1.0

¹BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

D. Reclamation Specifications:

1. The use of reclaimed water shall be limited to spray irrigation of native grasses or fodder, fiber, and seed crops and shall comply with the Department of Health Services' guidelines for the use of reclaimed water.

2. The use of reclaimed water must comply with the reclamation requirements of Title 22, Division 4, CCR (Section 60301 et seq.).
3. Reclaimed water shall not be applied to the irrigation areas during periods when soil is saturated.
4. Reclaimed water shall not be applied so as to cause saturated condition within 50 feet of any water body. There shall be no irrigation or impoundment of reclaimed water within 50 feet of any sump discharging into a surface drainage, or within 150 feet of any domestic well.
5. Treated effluent used for irrigation shall be managed to control runoff and prevent direct discharge to surface water. If tailwater is generated, it shall be returned to the ponds.
6. Discharge of reclaimed water shall be at agronomic rates consistent with good agricultural practices. Reclaimed water shall not be allowed to escape from the authorized use area by airborne spray or by surface flow.
7. Spray or runoff shall not enter a dwelling, food handling facility, nor any place where people may be present during irrigation.
8. The Discharger may not irrigate with effluent during periods of precipitation and for at least 24-hours after cessation of precipitation, or spray irrigate when wind velocities exceed 30 mph.
9. There shall be no cross-contamination between potable water supply and piping containing reclaimed water. Supplementing reclaimed water by connection with a domestic drinking water source or an irrigation or industrial well requires an air gap device or backflow prevention device that complies with Title 17, CCR, Section 7583, et seq.
10. All wastewater recycling equipment, pumps, piping, valves, and outlets shall be appropriately marked to differentiate them from potable facilities. The reclaimed water piping system shall not include any hose bibs.
11. The perimeter of the reclamation area shall be graded to prevent ponding along roads or other public areas. Perimeter warning signs indicating that reclaimed water is in use shall be posted at least every 500 feet, with a minimum of a sign at each corner of the reclamation area and at access road entrances.
12. Grazing of milking animals for human consumption within the area irrigated with reclaimed water is prohibited.

13. Reclaimed water shall be managed so as to prevent ponding and conditions conducive to the proliferation of mosquitoes and other disease vectors, and to avoid creation of a public nuisance or health hazard. More specifically,
 - a. Tailwater must be returned to the pond and all applied irrigation water must infiltrate or evaporate completely within 24 hours.
 - b. Ditches and low spots must be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.

E. Solids Disposal Requirements:

1. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. The Discharger shall submit disposal plans and sludge monitoring results to the Board at least **90 days** before any anticipated removal of sludge from any pond or the treatment plant. The Discharger may not discharge any sludge until the disposal plan is approved by the Executive Officer.
3. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
4. Use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
5. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger shall comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.
6. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*.

F. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

G. Surface Water Limitations:

The discharge shall not cause Indian Creek downstream of the percolation ponds to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

H. Provisions:

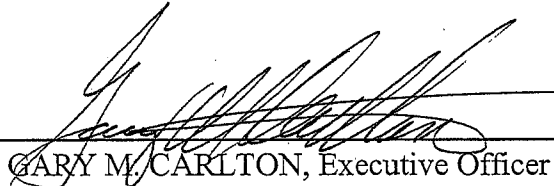
1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code.
 - a. By **28 December 2000**, the Discharger shall submit an Operation and Maintenance (O&M) Plan that describes procedures to operate the plant in compliance with this Order. The plan shall also address measures to be taken to remove vegetation from the pond dikes and within the ponds, and to also the control of cattail growth within Pond #1 if it is to be utilized as a wetland treatment cell. If the Discharger wishes cattails to remain in Pond #1, then the O&M Plan shall include a letter of approval from the local mosquito abatement district.
 - b. To determine compliance with the Groundwater Limitation, the Discharger shall submit a Groundwater Monitoring Workplan by **15 September 2001** and a Groundwater Well Installation Report within **120 days** after the approval of the workplan by the Executive Officer. The Groundwater Monitoring Workplan and Monitoring Well Installation Report shall be prepared by a Registered Geologist and shall contain the information listed in "*Items to be Included in a Monitoring Well Installation Report of Results.*"
 - c. By **15 May 2001**, the Discharger shall submit a water balance analysis and a long-term facility plan prepared by a Registered Engineer. The water balance shall calculate the monthly net flow volume into the system. Net flow shall include influent, inflow and infiltration, design seasonal precipitation based on a total annual precipitation using a return period of 100 years (distributed monthly in accordance with historical rainfall patterns), normal evaporation rates, and measured percolation rates. The water balance shall be used as a tool to develop a long-term facility plan. The plan shall address the storage, disposal, and water quality protection concerns of the facility. If the plan proposes modifications or additions to the current facility, such as utilization of the third pond and sprayfield, an engineering report and design plan shall be submitted with the long-term facility

plan. The report and plan shall (a) describe the design parameters of the modifications or additions, and (b) include a management and operations plan that will ensure that the system is operated in compliance with these WDRs.

- d. By **15 August 2001**, the Discharger shall have completed the proposed system modifications and/or additions described in the design plan.
 - e. **Disposal of biosolids** at any location other than Sonora Regional Wastewater Treatment Plant shall not be conducted without approval of the Executive Officer. If biosolids disposal is proposed at any other location, a biosolids disposal plan shall be submitted to and approved by the Executive Officer prior to the disposal. The biosolids disposal plan shall include biosolids monitoring results. The monitoring program shall be consistent with the biosolids monitoring prescribed in Monitoring and Reporting Program No. 5-00-201.
2. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-201, Which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
 3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
 4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
 5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
 6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
 7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.

8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 15 September 2000.


GARY M. CARLTON, Executive Officer

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-00-201
FOR
YOSEMITE VISTA ESTATES MUTUAL WATER AND SANITATION COMPANY
NOVASEL AND SCHWARTE INVESTMENT INC.
YOSEMITE VISTA ESTATES
TUOLUMNE

This Monitoring and Reporting Program (MRP) describes requirements for monitoring domestic wastewater, groundwater, and surface water. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

INFLUENT MONITORING

Samples shall be collected at the same frequency and at approximately the same time as effluent samples and should be representative of the influent for the sampling period. Influent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
BOD ₅ ¹	mg/l	grab	Monthly	Monthly
Total Settleable Solids	ml/l•hr	grab	Monthly	Monthly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

EFFLUENT MONITORING

Wastewater effluent samples shall be collected at the outlet before entering the percolation ponds. Grab samples are considered adequately composited to represent the effluent. Effluent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous	Daily	Monthly $\overline{Q} = 19 \text{ K}$
BOD ₅ ¹	mg/l	Grab	Weekly	Monthly $\overline{BOD} = 1.1$
Total Settleable Solids	ml/l•hr	Grab	Weekly	Monthly $\overline{SS} = 0.5$
pH	pH units	Grab	Weekly	Monthly $\overline{pH} = 7.5$
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly
Ammonia as Nitrogen	mg/l	Grab	Quarterly	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly	Quarterly
Total Coliform Organisms ²	MPN/100 ml	Grab	Quarterly	Quarterly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

² Using a minimum of 30 tubes or six dilutions

POLISHING POND MONITORING

Samples shall be collected from an established sampling station located in an area that will provide a sample representative of the wastewater in each pond. Freeboard monitoring shall be performed in both ponds. Freeboard will be measured vertically from the surface of the pond water to the lowest elevation of the surrounding berm. Freeboard shall be measured to the nearest 0.25 feet. Pond monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard	Feet	Measurement	Biweekly ¹	Monthly
Dissolved Oxygen	mg/l	Grab	Weekly	Monthly

¹ Biweekly shall mean twice per week.

GROUNDWATER MONITORING

Prior to construction of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed all new wells shall be added to the MRP, and shall be sampled and analyzed according to the schedule below.

Prior to sampling the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency²</u>
Groundwater elevation	Feet	Measurement	Quarterly
pH	pH units	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Electrical Conductivity	umhos/cm	Grab	Quarterly
Ammonia as Nitrogen	mg/l	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms ²	MPN/100 ml	Grab	Quarterly

¹If the criteria are exceeded, the Discharger shall immediately resample the groundwater.

² Using a minimum of 15 tubes or three dilutions

SURFACE WATER MONITORING

The Discharger shall establish two sampling stations along the north bank of Indian Creek: one station (S-1) shall be located 100 feet upstream of the easternmost percolation pond and one station (S-2) approximately 50 feet downstream of the westernmost percolation pond and sewage force main transfer line. Surface water samples shall be collected on the same day(s) as groundwater samples. Samples of the creek water shall be analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
pH	pH units	Grab	Quarterly
BOD ₅ ¹	mg/l	Grab	Quarterly
Dissolved Oxygen	mg/l	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Electrical Conductivity	umhos/cm	Grab	Quarterly
Ammonia as Nitrogen	mg/l	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms ²	MPN/100 ml	Grab	Quarterly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

² Using a minimum of 30 tubes or six dilutions

WATER SUPPLY MONITORING

The Discharger shall obtain and analyze a representative sample of the drinking water supply well as listed below. The results shall be presented in the Annual Summary Monitoring Report.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Standard Minerals	mg/l	Grab	Annually
PH	pH units	Grab	Annually
Total Dissolved Solids	mg/l	Grab	Annually
Electrical Conductivity	umhos/cm	Grab	Annually
Nitrates as Nitrogen	mg/l	Grab	Annually
Total Coliform Organisms	MPN/100 ml	Grab	Annually

BIOSOLIDS MONITORING

If disposal of biosolids at any other location other than Sonora Regional Wastewater Treatment Plant is proposed, a minimum of one composite sample of biosolids shall be collected and results submitted along with a biosolids disposal plan. The sample shall be tested for the following constituents:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Total Solids	Percent	Grab	Annually
Heavy Metals ^{1,2}	mg/Kg	Grab	Annually
Fecal Coliform Organisms ²	MPN/100 ml	Grab	Annually

¹ Arsenic, Cadmium, Copper, Chromium, Lead, Mercury, Molybdenum, Nickel, Selenium, and Zinc.

² To be reported on a dry weight basis.

For all biosolids generated, a log shall be kept that document biosolids quantities generated; sampling dates and analytical reports, if sampling and analysis is conducted; and handling, storage, and disposal practices. This information shall be submitted to the Regional Board as part of the Annual Monitoring Report.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, and polishing ponds monitoring.
2. A statement describing conditions around the percolation ponds relative to the amount of vegetation.
3. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format.
4. If requested by staff, copies of laboratory analytical report(s).
5. A calibration log verifying weekly calibration of all monitoring instruments and devices used to fulfill the prescribed monitoring program.

B. Quarterly Monitoring Reports

The Discharger shall establish a quarterly sampling schedule for groundwater, surface water and effluent monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Board by the **1st day of the second month after the quarter** (i.e. the January-March quarter is due by May 1st) each year. The Quarterly Report shall include the following:

1. Results of groundwater, surface water, and effluent monitoring. The results of regular monthly monitoring reports for March, June, September and December may be incorporated into their corresponding quarterly monitoring report.
2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for the groundwater and surface water monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. Field logs shall support the narrative for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged.
3. Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any.

4. A narrative discussion of the analytical results for all media and locations monitored including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable).
5. A comparison of monitoring data to the discharge specifications, groundwater limitations and surface water limitations, and explanation of any violation of those requirements.
6. Summary data tables of historical and current water table elevations and analytical results.
7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and any other sampling stations, and groundwater elevation contours referenced to mean sea level datum.
8. Copies of laboratory analytical report(s) for groundwater and surface water monitoring.

C. Annual Report

An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February** of each year. In addition to the data normally presented, the Annual Report shall include the following:

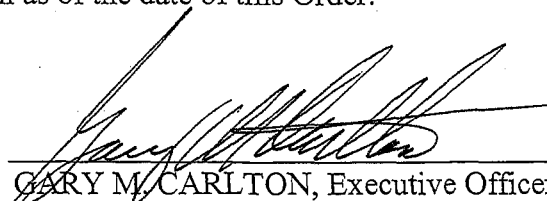
1. The contents of the regular quarterly monitoring report for the last quarter of the year.
2. If requested by staff, tabular and graphical summaries of all data collected during the year.
3. Data for monitoring and analysis performed on an annual basis (i.e., biosolids and water supply monitoring).
4. An evaluation of the performance of the domestic wastewater treatment system and evaporation/percolation ponds, as well as a forecast of the flows anticipated in the next year.
5. An evaluation of the groundwater quality beneath the wastewater treatment facility.
6. Summary of information on the disposal of all biosolids including volume removed and location of disposal.
7. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.
8. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

MONITORING AND REPORTING PROGRAM NO. 5-00-201
YOSEMITE VISTA ESTATES
TUOLUMNE COUNTY

-7-

The Discharger shall implement the above monitoring program as of the date of this Order.

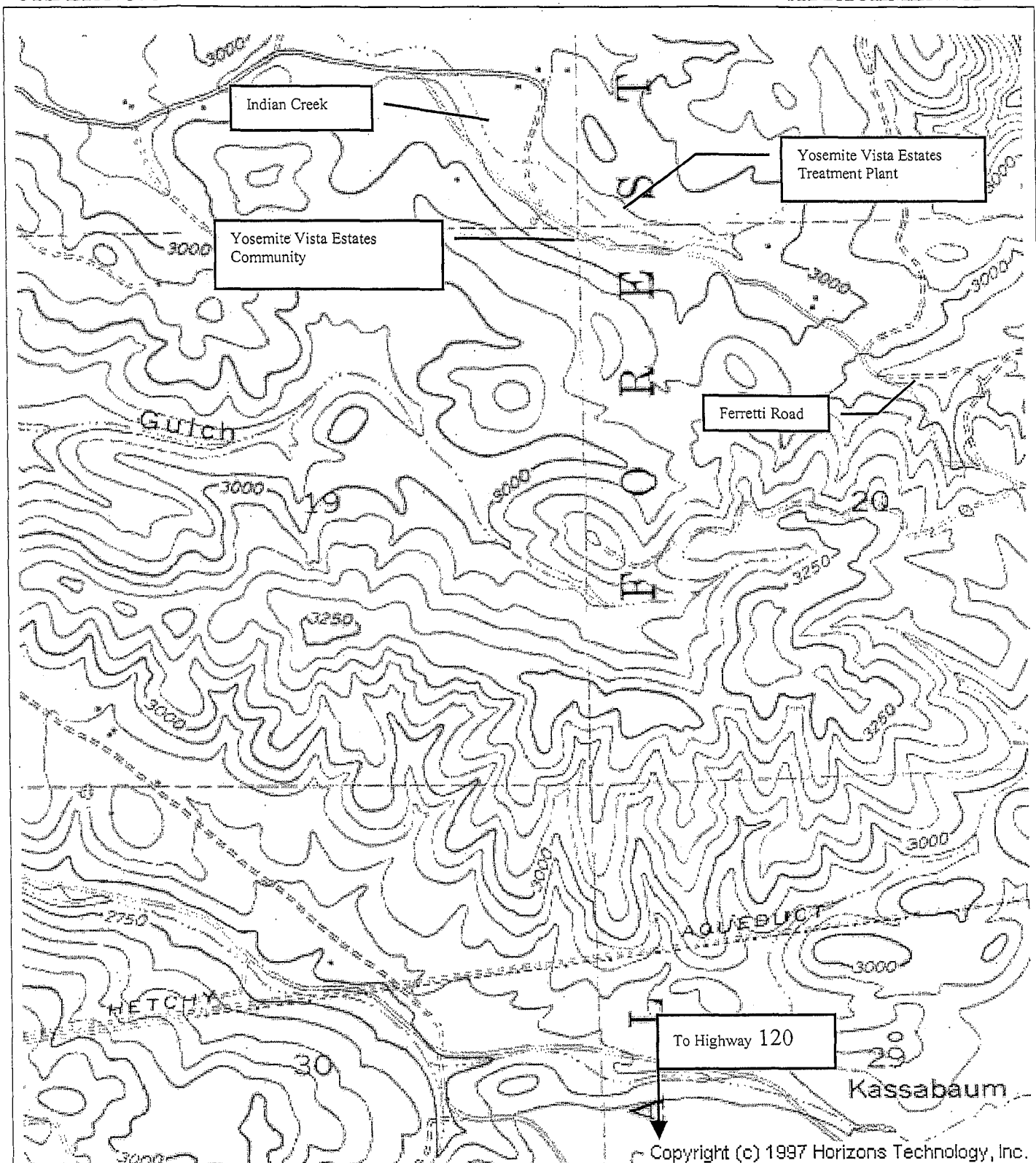
Ordered by:


GARY M. CARLTON, Executive Officer

15 September 2000

(Date)

JRM 9/15/00

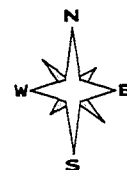


Drawing Reference:

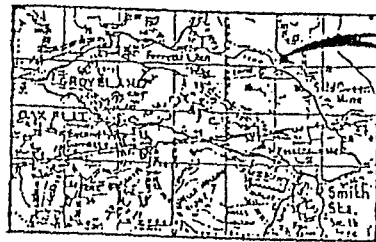
Lake Eleanor
U.S.G.S TOPOGRAPHIC MAP
7.5 MINUTE QUADRANGLE
Photorevised 1956

SITE LOCATION MAP

Yosemite Vista Estates Mutual Water And Sanitation
Company
Novasel and Schwarte Investment Inc.
Yosemite Vista Estates
Tuolumne County
Section 20, T1S, R17E, MDB&M



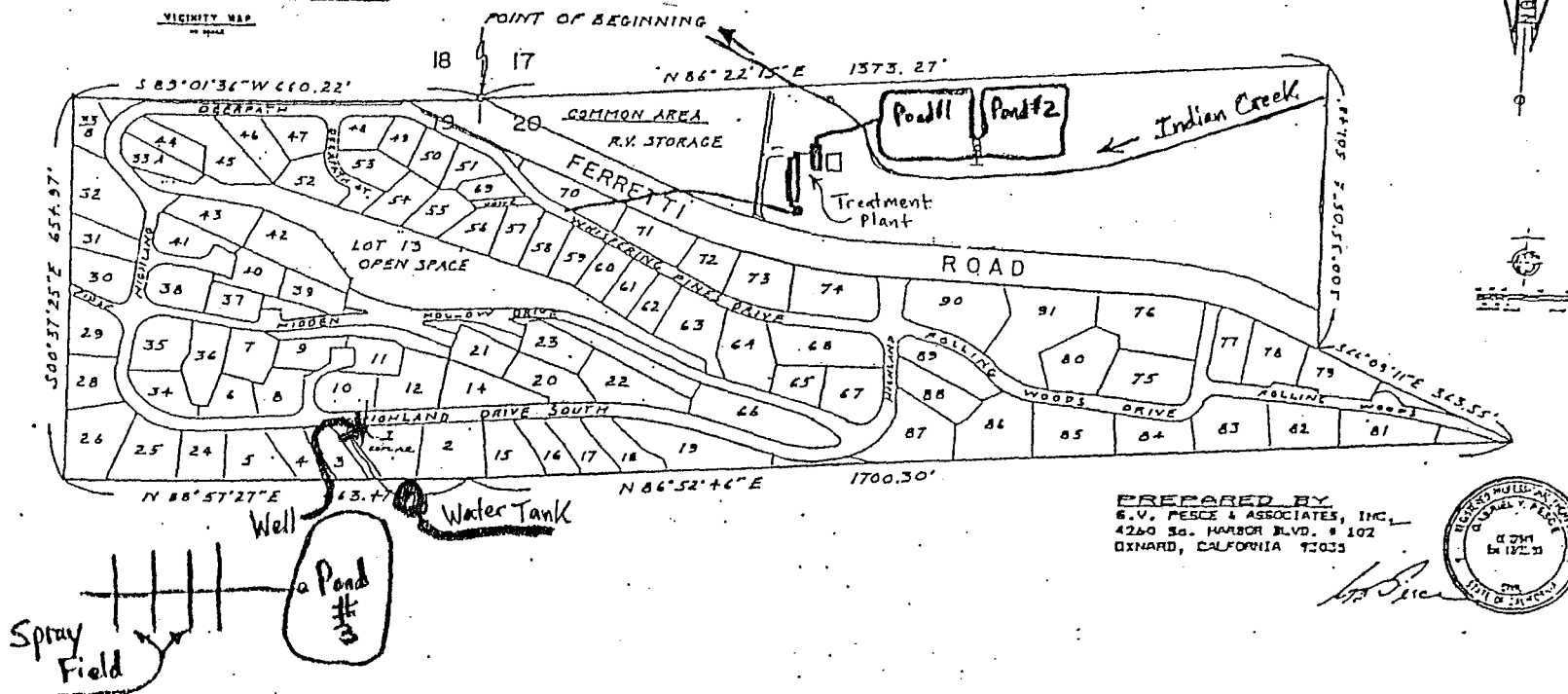
approx. scale
1 in. = 1,350 ft.



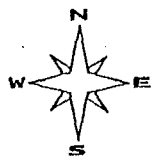
VICINITY MAP

SITE MAP

YOSEMITE VISTA ESTATES



PREPARED BY
G.V. PESCE & ASSOCIATES, INC.
4260 SO. HARBOR BLVD. # 102
OXNARD, CALIFORNIA 93033



Drawing Reference:
Yosemite Vista Estates
G.V. Pesce & Associate, Inc.
4260 So. Harbor Blvd. #102
Oxnard, Ca

SITE PLAN
Yosemite Vista Estates Mutual Water And Sanitation Company
Novasel and Schwarte Investment Inc.
Yosemite Vista Estates
Tuolumne County
Section 20, T1S, R17E, MDB&M



California Regional Water Quality Control Board

Central Valley Region

Steven T. Butler, Chair



Gray Davis
Governor

Winston H. Hickox
Secretary for
Environmental
Protection

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ATTACHMENT C WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00201

ITEMS TO BE INCLUDED IN A MONITORING WELL INSTALLATION WORKPLAN AND A MONITORING WELL INSTALLATION REPORT OF RESULTS

Prior to installation of groundwater monitoring wells, the Discharger shall submit a workplan containing the minimum listed information. Wells may be installed after staff approve the workplan. Upon installation of the monitoring wells, the Discharger shall submit a report of results, as described below. All workplans and reports must be signed by a registered geologist, certified engineering geologist, or civil engineer registered or certified by the State of California.

Monitoring Well Installation Workplan

A. General Information:

- Monitoring well locations and rationale
- Survey details
- Equipment decontamination procedures
- Health and safety plan
- Topographic map showing any existing monitoring wells, proposed wells, waste handling facilities, utilities, and other major physical and man-made features.

B. Drilling Details: describe drilling and logging methods

C. Monitoring Well Design:

- Casing diameter
- Borehole diameter
- Depth of surface seal
- Well construction materials
- Diagram of well construction
- Type of well cap
- Size of perforations and rationale
- Grain size of sand pack and rationale
- Thickness and position of bentonite seal and sand pack
- Depth of well, length and position of perforated interval

D. Well Development:

- Method of development to be used
- Method of determining when development is complete
- Method of development water disposal

- E. Surveying Details: discuss how each well will be surveyed to a common reference point
- F. Soil Sampling (if applicable):
 - Cuttings disposal method
 - Analyses to be run and methods
 - Sample collection and preservation method
 - Intervals at which soil samples are to be collected
 - Number of soil samples to be analyzed and rationale
 - Location of soil samples and rationale
 - QA/QC procedures
- G. Well Sampling:
 - Minimum time after development before sampling (48 hours)
 - Well purging method and amount of purge water
 - Sample collection and preservation method
 - QA/QC procedures
- H. Water Level Measurement:
 - The elevation reference point at each monitoring well shall be within 0.01 foot. Ground surface elevation at each monitoring well shall be within 0.1 foot. Method and time of water level measurement shall be specified.
- I. Proposed time schedule for work.

Monitoring Well Installation Report of Results

- A. Well Construction:
 - Number and depth of wells drilled
 - Date(s) wells drilled
 - Description of drilling and construction
 - Approximate locations relative to facility site(s)
 - A well construction diagram for each well must be included in the report, and should contain the following details:
 - Total depth drilled
 - Depth of open hole (same as total depth drilled if no caving occurs)
 - Footage of hole collapsed
 - Length of slotted casing installed
 - Depth of bottom of casing
 - Depth to top of sand pack
 - Thickness of sand pack
 - Depth to top of bentonite seal
 - Thickness of bentonite seal
 - Thickness of concrete grout
 - Boring diameter

Casing diameter
Casing material
Size of perforations
Number of bags of sand
Well elevation at top of casing
Depth to ground water
Date of water level measurement
Monitoring well number
Date drilled
Location

B. Well Development:

Date(s) of development of each well
Method of development
Volume of water purged from well
How well development completion was determined
Method of effluent disposal
Field notes from well development should be included in report.

C. Well Surveying: provide reference elevations for each well and surveyor's notes

D. Water Sampling:

Date(s) of sampling
How well was purged
How many well volumes purged
Levels of temperature, EC, and pH at stabilization
Sample collection, handling, and preservation methods
Sample identification
Analytical methods used
Laboratory analytical data sheets
Water level elevation(s)
Groundwater contour map

E. Soil Sampling (if applicable):

Date(s) of sampling
Sample collection, handling, and preservation method
Sample identification
Analytical methods used
Laboratory analytical data sheets

INFORMATION SHEET

WASTE DISCHARGE REQUIREMENTS NO. 5-00-201
YOSEMITE VISTA ESTATES MUTUAL WATER AND SANITATION COMPANY
NOVASEL AND SCHWARTE INVESTMENT INC.
YOSEMITE VISTA ESTATES
TUOLUMNE COUNTY

The Yosemite Vista Estates is located approximately 7.5 miles east of Groveland on Ferretti Road. The treatment plant is also located on Ferretti Road and is immediately adjacent to Indian Creek. Wastewater originates from domestic sources in a 28-acre residential subdivision including a clubhouse and pool.

The flow rates are expected to increase over time with growth in the development. The current estimated flow rate of 5,000 – 6,000 gpd is projected to increase to around 12,000 gpd by the year 2005, and to 19,000 by 2010.

Sewage is conveyed from the residential sites to the treatment plant through a gravity collection system. The plant treatment units consist of an aeration tank, including contact chamber of 22,500 gallons, and a secondary clarifier of 6,000 gallons. The treatment plant is rated for a total daily flow of 21,500 gpd with an influent daily loading rate of 45 pounds of BOD.

Effluent from the activated sludge package treatment plant is pumped from a 750 gallon stilling well through a 2.5 inch force main into first of two percolation pond. The ponds are connected by a channel and are operated in series fashion. The estimated storage capacity of the ponds is 400,000 gallons for the first pond and 570,000 gallons for the second pond. The ponds are located within 30 feet of Indian Creek.

Surface water monitoring of Indian Creek was not required under the previous WDRs. The impact of the percolation ponds in such close proximity to Indian Creek is not known at this time. In addition, the impacts of the percolation ponds on the underlying groundwater are not known. Therefore, surface water and groundwater monitoring are required as part of this Order. A third pond and sprayfield are located on the south side of the development and were authorized for use under the previously WDRs, but never utilized. The Discharger has proposed to incorporate the pond, along with the sprayfield, into the treatment/disposal process at a future date to help accommodate growth should the capacity of the current pond system become exceeded. The Discharger did not include specifications for the use of the spray field with the April 2000 RWD. Therefore, this Order, although authorizing the use of the third pond and sprayfield for the disposal of wastewater, does so only after submittal of an engineering report and design plan, and approval of these reports by the Executive Officer.

Waste sludge generated from the treatment plant is generally hauled to the Regional Wastewater Treatment Plant in Sonora for disposal. However, the Discharger also dries a small amount of sludge on site and applies it as compost to the land around the treatment plant. This Order requires a biosolids disposal plan and monitoring results if the Discharger wishes to continue applying sludge on site.

